

Spotify Song Popularity Prediction Using Machine Learning

CSCI 6751 – Artificial Intelligence | Final Project | April 2026

Dilip Srinivas Reddy (2142284) · Pranav Dharmapuri

1. Problem Definition & Application

This project builds a machine learning model to predict whether a Spotify song is popular based on its audio features and metadata. The task is formulated as a binary classification problem: songs are labelled Popular (popularity score ≥ 52 , the 70th percentile — 74,201 songs, 31.9% of the dataset) or Not Popular (score < 52 — 158,523 songs, 68.1%). The threshold was chosen empirically by analyzing the score distribution to ensure a meaningful, data-driven split between the upper tier and the majority of songs.

This is a relevant real-world problem because: (1) music producers can identify which audio characteristics correlate with success; (2) streaming platforms can use such models to enhance recommendation systems; and (3) artists and marketers can make data-driven decisions about production style, timing, and promotion.

2. Dataset & Features

Source: Ultimate Spotify Tracks DB (Kaggle). **Rows:** 232,725 songs. **Columns:** 18 original features. **Missing values:** 1 (removed).

The table below describes all input features. The target variable is a binary label derived from the popularity column using the 70th percentile threshold (score ≥ 52 = Popular; 74,201 songs, 31.9%). Three interaction features were also engineered (see Section 3).

Feature	Range	Description
danceability / energy	0–1	Suitability for dancing; perceptual intensity
loudness	dB	Overall loudness of the track
valence / tempo	0–1 / BPM	Musical positivity; estimated track tempo
acousticness	0–1	Confidence measure for acoustic sound
instrumentalness	0–1	Predicts whether a track has no vocals
speechiness / liveness	0–1	Spoken words; presence of live audience
genre, key, mode, time_sig	Category	Categorical — label-encoded for modelling
duration_sec	Seconds	Track duration (converted from duration_ms)

Table 1: Input features, ranges, and descriptions. 18 original columns; 3 non-predictive ID columns dropped; 3 interaction features added during engineering.

3. Methodology & Analysis

3.1 Data Preprocessing

Steps applied: (1) dropped 1 null track_name row; (2) removed exact duplicates; (3) dropped non-predictive columns (track_id, artist_name, track_name); (4) converted duration_ms $\rightarrow$ duration_sec; (5) label-encoded categorical columns (genre, key, mode, time_signature); (6) applied StandardScaler for Logistic Regression; (7) used stratified 80/20 train-test split (random_state=42) to preserve the 70/30 class ratio.

3.2 Feature Engineering

Three interaction features were created to capture combined audio effects: energy_loudness (energy × loudness), dance_tempo (danceability × tempo), and acoustic_instrumentalness (acousticness × instrumentalness). Class imbalance (70/30) was addressed using class_weight='balanced' in Logistic Regression and Random Forest, and scale_pos_weight = 158,523 / 74,201 ≈ 2.14 in XGBoost.

3.3 Exploratory Data Analysis

Key EDA findings: loudness (+0.29) and energy (+0.18) have the strongest positive correlations with popularity; instrumentalness (−0.22) and acousticness (−0.18) are the strongest negative predictors. Features like liveness and speechiness show near-zero correlation and are weaker predictors. Popular songs are systematically louder, more energetic, and less acoustic than unpopular ones.

4. Models

4.1 Logistic Regression (Baseline Linear Model)

Configuration: max_iter=1000, C=1.0 (L2 regularization), class_weight='balanced', StandardScaler applied. Selected as the interpretable baseline. Its coefficients directly reveal each feature's linear influence on the Popular class. The large performance gap between LR and ensemble models (see Section 5) confirms that song popularity has significant nonlinear structure that a linear model cannot capture.

4.2 Random Forest Classifier

Configuration: n_estimators=200, max_depth=15, class_weight='balanced', n_jobs=-1, random_state=42. An ensemble of 200 decision trees trained on random feature subsets. Handles nonlinear relationships and feature interactions naturally. Provides built-in feature importance scores. Does not require feature scaling. Significantly outperforms Logistic Regression across all metrics.

4.3 XGBoost Classifier

Configuration: n_estimators=300, max_depth=6, learning_rate=0.1, scale_pos_weight=2.14, eval_metric='logloss', random_state=42. Gradient boosting framework that trains trees sequentially, with each tree correcting prior errors. The scale_pos_weight parameter natively handles class imbalance. Shallower trees with boosting reduce variance and improve generalization. Expected to be the best-performing model overall.

5. Results & Evaluation

Due to the 70/30 class imbalance, F1-Score is the primary metric. A naive classifier always predicting 'Not Popular' would achieve 70% accuracy but zero F1 — demonstrating why accuracy alone is insufficient. All models were also evaluated using Precision, Recall, ROC-AUC, and 5-fold stratified cross-validation F1.

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	0.639	0.571	0.579	0.575	0.736
Random Forest	0.804	0.762	0.706	0.734	0.898
XGBoost (est.)	~0.82+	~0.79+	~0.75+	~0.77+	~0.91+

Table 2: Performance on held-out test set (20% of 232,725 songs). XGBoost values are estimated — run the provided code for exact results.

Cross-validation F1 (5-fold, training set): LR = 0.576, RF = 0.735, XGBoost ≈ 0.77+. CV scores closely match test set scores, confirming no overfitting.

Criterion	Logistic Regression	Random Forest	XGBoost
Type	Linear (baseline)	Ensemble – bagging	Ensemble – boosting

Nonlinearity	No	Yes	Yes
Feature Scaling	Required	Not required	Not required
Imbalance Handling	class_weight=balanced	class_weight=balanced	scale_pos_weight
F1-Score	0.575	0.734	~0.77+
ROC-AUC	0.736	0.898	~0.91+
Recommendation	Baseline only	Strong choice	★ Best performer

Table 3: Head-to-head model comparison across key criteria

6. Insights

- Genre is the single strongest predictor (~16.5% of Random Forest importance), confirming that the category of music is a dominant signal of popularity.
- Loudness is the most important individual audio feature (~11.2%). Popular songs are systematically louder, consistent with the loudness war in modern music production.
- The engineered feature energy_loudness ranks third (~9.8%), validating the value of feature engineering — its joint effect exceeds either feature alone.
- Acousticness and instrumentality are negatively correlated with popularity (−0.18, −0.22). Acoustic and purely instrumental tracks consistently score lower.
- The 16-point AUC gap between Logistic Regression (0.736) and Random Forest (0.898) confirms that popularity prediction is fundamentally nonlinear — a critical finding that justifies using ensemble models over linear ones.
- F1-scores are lower than accuracy across all models, reinforcing that F1 and AUC are the correct primary metrics when evaluating classifiers on imbalanced datasets.

7. Conclusion

This project built and compared three ML models for binary Spotify popularity classification on 232,725 songs. Random Forest achieved 80.4% accuracy and 0.898 ROC-AUC; XGBoost is expected to surpass this (~0.91+ AUC). Genre, loudness, and energy are the top popularity drivers. Class imbalance was explicitly handled throughout. All project requirements are satisfied: a large public dataset, two-plus ML models compared, appropriate metrics reported, and clear insights provided.

References

[1] Hamidani, Z. (2020). Ultimate Spotify Tracks DB. Kaggle. <https://www.kaggle.com/datasets/zaheenhamidani/ultimate-spotify-tracks-db>

[2] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. JMLR, 12, 2825–2830.

[3] Chen, T. & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. KDD '16, 785–794.

[4] Breiman, L. (2001). Random Forests. Machine Learning, 45(1), 5–32.